



Strategic investment planning for the hydrogen economy – A mixed integer non-linear framework for the development and capacity expansion of hydrogen supply chain networks

Salvador I. Pérez-Uresti^{a,*}, Gustavo Gallardo^a, Dimitrios K. Varvarezos^b

^a Aspen Technology Inc., Corporate Research Group, Av. Paseo de la reforma 412, Suite 8, Cuauhtémoc, MX, CDMX 06600, Mexico

^b Aspen Technology Inc., Corporate Research Group, 2500 City West Blvd., Suite 1800, Houston, TX 77042, USA

ARTICLE INFO

Keywords:

Hydrogen economy
Carbon capture
Optimization
Supply chain
Capacity expansion
Strategic investment planning

ABSTRACT

This work presents a novel Mixed-Integer Nonlinear Programming (MINLP) modeling and optimization framework for the investment planning of large-scale hydrogen economy envelopes. The scope of the model includes the design, synthesis, and long-term capacity expansion of hydrogen supply chain networks (HSCN) that include hydrogen production, purification, storage, transportation, and distribution, subject to environmental sustainability considerations. Production and carbon capture units are represented using surrogate models derived from rigorous simulations to capture important process non-linearities. This framework provides an optimal roadmap for the 10-year plan of hydrogen infrastructure development in the state of California, that demonstrates the important trade-offs between investment decisions, economic incentives, and regulatory carbon emissions constraints. It is shown that the development of the hydrogen production and distribution networks can be achieved over a 10-year horizon in an economically profitable way, while achieving a 90 % reduction in carbon emissions and satisfying all state regulatory mandates.

1. Introduction

The use of hydrogen as an energy carrier is being prompted nowadays by the growing interest in finding solutions that can contribute to the net zero emissions goal set by 2050. Hydrogen has a high-energy content, a low-carbon intensity, and it can be obtained from a wide range of low-carbon sources (IEA, 2019). Current applications of hydrogen are mostly industrial, having only a very small footprint in the transportation area. Nevertheless, the hydrogen economy is expected to expand rapidly in the next decade. The report elaborated by Hydrogen Council (2021), identified up to 35 new applications in transportation, heating, and power sectors out of which 22 will become cost-competitive by 2030. As a result, it is expected that hydrogen will make up an 18 % share of the total final energy consumption by 2050 (International Renewable Energy Agency IRENA, 2019). In this new context, decisions makers will invest in solutions that address the future challenges related to the entire hydrogen supply chain including production, storage, transportation, and end-use cases.

Fossil fuels are the primary sources of hydrogen with a 79 % share of the global production. Steam methane reforming (SMR) and coal

gasification are currently the most common production technologies. The production of hydrogen in 2020 resulted in 900 Mt of CO₂ emissions, equivalent to the emissions of Indonesia and the United Kingdom combined (IEA, 2021). Further efforts are needed to offer cost-effective and cleaner alternatives to produce hydrogen. In this context, some countries in North America and Europe have been deploying projects accounting for carbon capture technologies in hydrogen production (Yu et al., 2021).

Alternative routes have also been developed including the use of renewable resources as raw materials with no-direct emissions generated. In this respect, electrolysis is nowadays the most promising of those technologies. Increasingly, hydrogen producers are leading projects to deploy assets incorporating renewables-based electrolysis to produce hydrogen with minimal carbon footprint. Some examples are the projects led by Shell (2022); (Rephyne, 2021), Iberdrola (2022) and Linde (2022). Even more, Guerras et al. (2019) reported that electrolysis could produce hydrogen at competitive prices against the fuel cost of gasoline vehicles when dynamic pricing for electricity is used in some states of the US. On the other hand, hydrogen can also be obtained from biomass through a gasification process; however, the production costs are still higher when compared with conventional process (Davis and

* Corresponding author.

E-mail address: salvador.perez@aspentech.com (S.I. Pérez-Uresti).

Nomenclature**Sets**

I	hydrogen production technologies ($\{i i = \text{SMR, CG, ...}\}$)
J	flue gas components ($\{j j = \text{CO}_2, \text{N}_2, \text{O}_2\text{H}_2\text{O}\}$)
Z	zones/customers ($\{z z = 1, \dots, N\}$)
T	periods of time ($\{t t = 1, \dots, N\}$)
K	raw material ($\{k k = \text{Nat. gas, Coal, ...}, \text{Biomass}\}$)
F	quality of hydrogen produced ($\{f f = \text{low, high}\}$)
L	transportation mean ($\{l l = \text{LT, GT, PIP}\}$)
ZZ'	transfers conducted between two zones ($\{z \rightarrow z' z \rightarrow z' = 1, \dots, Z\}$)
S	types of hydrogen storage ($\{s s = \text{LH, GH}\}$)

Parameters

$R_{i,k}^{\text{HYD}}$	tons of H_2 produced in unit i per tons of raw material k
$R_{i,k}^{\text{CRD}}$	tons of CO_2 produced in unit i per tons of raw material k
$R_{i,k}^{\text{KWH}}$	MWh of electricity required by unit i per tons of raw material k
$A_{i,t}^r$	cost parameter depending on the size of plant (k\$/ton)
$B_{i,t}^r$	cost constant for technology I (k\$)

Integer variables

$y_{i,z,t}^I$	binary variable used to denote the installation of technology i at period t
$y_{i,z,t}^{\text{re}}$	binary variable used to denote the retirement of technology i at period t
$N_{i,z,t}^r$	number of plants installed at each time-period and technology i
$N_{i,z,t}^{\text{re}}$	number of plants retired at each time-period and technology i
$y_{s,f,z,t}^S$	binary variable indicating the installation of storage type s in zone z for product f
$y_{f,z \rightarrow z',t}^T$	binary variable denoting the existence of hydrogen transfer from zone z to zone z'
$N_{f,l,t}^{\text{TRA}}$	total number of trucks/pipelines installed or purchased at each period of time.

Continuous variables

$F^{U1,U2}$	mass flowrate of stream U1 at unit U2 (t/d)
$F_j^{U1,U2}$	mass flowrate of component j in stream U1 at unit U2 (t/d)
$x_j^{U1,U2}$	mass fraction of component j in stream U1 at unit U2 (t/d)
$F_{k,z,i,t}$	raw material k fed to unit i at zone z (t/d)
$H_{i,z,t}^{\text{GEN}}$	hydrogen generated by technology I at zone z in time period t (t/d)
$KWH_{z,i,t}$	electricity consumed by technology I at zone z in time period t (MWh)
$CRD_{z,i,t}$	hydrogen generated by technology I at zone z in time period t (t/d)
$H_{i,f,z,t}^{\text{PU}}$	hydrogen coming from the purification facilities and generated by technology i (t/d)
$H_{f,z \rightarrow z',t}^{\text{TRA}}$	hydrogen imported from neighboring zones z' (t/d)
$H_{s,f,z,t}^{\text{DIS}}$	hydrogen discharged by storage s in zone z (t/d)
$H_{s,f,z,t}^{\text{CHA}}$	hydrogen charged in storage s in zone z (t/d)
$Cap_{i,z,t}$	production capacity of technology I at zone t (t/d)
$Cexp_{i,z,t}^r$	expanded capacity r at period t (t/d)
$Cre_{i,z,t}^{\text{re}}$	capacity retired at period t , technology i (t/d)
$H_{s,f,z,t}^{\text{STO}}$	hydrogen stored in storage s at period t (ton)
$Cap_{z,t}^{\text{rs}}$	fueling station capacity rs in zone z at period t (t/d)
$Cexp_{z,t}^{\text{rs}}$	expanded capacity of fueling station rs in zone z at period t (t/d)
$CI_{i,z,t}$	amortized capital investment cost of technology I at period t (k\$/yr)
$CIPur_{i,z,t}$	investment purification cost for hydrogen f generated by technology i (\$MM)
$S_{\text{off-gas } i,z,t}$	off-gas flowrate produced by technology I (t/d)
$TC_{f,l,t}$	amortized transportation cost (k\$/yr)
$N_{f,l,z \rightarrow z',t}$	average number of trucks transferring hydrogen between two zones
$RSC_{z,t}$	amortized refueling station cost (k\$/yr)
$FC_{f,l,z \rightarrow z',t}$	cost of fuel for transferring hydrogen using trucks between two zones

Martin 2014). Mosca et al. (2020) evaluated the use of bioethanol as raw material to obtain hydrogen through a steam ethanol reforming. Although the production cost yielded twice the one of conventional SMR, the authors claimed that a reduction of 65 % of carbon emissions can be reached.

A more diversified hydrogen economy will also include specialized applications requiring different purity standards. Du et al. (2021) report that up to three purity grades of hydrogen are being used in the transportation sector ranging from pure (99.9 %) to ultra-pure (99.999 %). Under this scheme, the purity costs not only will depend on the technology used to produce hydrogen but also on its final application. This item was addressed by Wickham et al. (2022) who concluded that the purity cost could represent as much as 30 % of the total discounted investment cost of a hydrogen supply chain designed for UK. The authors proposed the use of correlations to estimate the purification cost based on the initial and final hydrogen purity. The hydrogen purity obtained from a steam methane process was assumed to be 95 %. Therefore, additional purification is needed so that hydrogen can be sent to the fueling stations. In this work, we present an approach that addresses two distinct purification steps. The first one is calculated based on the product distribution obtained from each type of technology and the second is used to adjust the purity to the transportation specifications when required.

Modeling hydrogen production represents only a fraction of the

whole supply chain. The complete Hydrogen Supply Chain Network (HSCN) also includes hydrogen transportation and storage. Hydrogen requires high transportation and storage volumes because of its low volumetric energy density (68 % lower than natural gas), which increases the final cost. The best choice partially relies on the scale of HSCN. For instance, Yang and Ogden (2007) reported that when an amount lower than 14 MT/day is transported over a short distance, the use of gas trucks becomes the most economical option. For longer distances, the use of liquid hydrogen trucks and pipelines is preferred. Other solutions in this respect consider the conversion of hydrogen into compounds with higher volumetric energy density, i.e., ammonia and liquid organic carriers (Hydrogen Council, 2021; Chyoda Corporation, 2020) and blending it into natural gas pipelines (Melania et al. 2013; Penchev et al., 2022). One example of the latter is the HyNTS project led by the National Grid in UK, which aims at the full conversion of the gas transmission system to transport hydrogen. (Sadler et al., 2016; National Grid, 2021). The challenge for both options is that the H_2 recovery step at the point of use requires a large amount of energy.

Storage systems of hydrogen are essential in the operation of the HSCN (Hydrogen Supply Chain Network) and must be flexible enough to synchronize supply and demand. The most common storage means are geological storage as well as compressed and liquefied gas vessels. The high diffusion of hydrogen in solids, boil-off ratio, and cushion gas are the most relevant parameters to be considered when selecting a suitable

storage technology. For instance, [Tarkowski \(2019\)](#) point out that due to the low hydrogen diffusion in water, the depleted hydrocarbon deposits and aquifers would be preferred over the salt caverns, which are usually dry structures. A large amount of hydrogen can be stored using liquefaction technology; however, the high energy intensity and the high boil-off rates are drawbacks for long-term storage ([Barthelemy et al., 2017](#)).

The optimal design of a HSCN has been traditionally addressed by formulating a mixed-integer linear programming (MILP) model that includes a superstructure representation of the alternative production, storage, transmission, as well as primary and secondary distribution modes and technologies. The work by [Almansoori and Shah \(2006\)](#) presented a modeling framework to include production, transport, and storage of hydrogen with no explicit time consideration. This model was then extended in [Almansoori and Shah \(2009\)](#) to a multi-period approach by adding hydrogen demand variations, availability of energy resources, capacity expansion, fueling stations, and local and regional transfers. [Guillén-González et al. \(2010\)](#) addressed the problem using a multi-objective optimization approach where life cycle assessment (LCA) principles were used to build a second objective function aimed at reducing the environmental impact of the HSCN. A similar approach was used by [Sabio et al. \(2010\)](#) who considered the financial risk, driven by uncertain operating cost, as a third objective function. [Agnolucci et al. \(2013\)](#) incorporated spatially detailed scenarios for hydrogen demand, which are estimated using socioeconomic data including access to cars, education, commuting distance, and household income. The model also includes decisions on the infrastructure needed for the final disposition of CO₂ when carbon capture and storage (CCS) is selected. [Reuß et al. \(2019a\)](#) also addressed the spatial distribution of demand with a special focus on hydrogen transmission and distribution blocks. As part of the transportation modes considered, the model incorporated the use of hydrogen carriers such as liquid organics (LOC). The authors concluded that under restrictive scenarios of CO₂ emissions, the LOCs were not competitive compared to conventional transports because of the heat required to conduct the dehydrogenation step at the fueling station. [Nunes et al. \(2015\)](#), formulated a stochastic model to account for the uncertainty of hydrogen demand over time. First-stage decisions were investments in production plants and storage facilities, while second-stage decisions were transportation contracts and supply chain operation logistics. Other examples of models considering uncertainty include the ones developed by [Almansoori and Shah \(2012\)](#) and [Bique et al. \(2021\)](#).

Despite the extend of the cited research, none of the models mentioned above capture the intermittent behavior of renewables over time, nor do they consider the interactions of the power grid with hydrogen production. Both items are of interest as the share of green hydrogen is expected to increase. To address this limitation, [Samsatli et al. \(2016\)](#) proposed a model that considers operational and design decisions of hydrogen supply chains interacting with wind farms. Under this scheme, energy can be transmitted as electricity and hydrogen. The model includes the characterization of the hourly variability of wind velocity, and the spatial-temporal characterization of hydrogen demand using Geographic Information Systems (GIS). This work was extended by [Samsatli and Samsatli \(2019\)](#) to include household heating demand in the UK. Here hydrogen is sent directly to domestic combined heat-power plants or to hydrogen boilers, which generate high-grade heat used in district heating. Results showed that over 20 % of heat could be covered by hydrogen. [He et al. \(2021\)](#) modeled the interactions with the grid assuming distinct scenarios of power production mix under different carbon prices. The authors conclude that a higher carbon price leads to a higher share of electrolysis in hydrogen production. The dynamics of storage and transmission systems were also affected by the electricity price. During the off-peak electricity price periods and low demands for hydrogen, the system increases the hydrogen stored that is then released during the high hydrogen demand periods. Some authors have considered the rigorous modeling of pipeline flows ([Reuss et al.](#)

[2019b](#)) and hydrogen secondary purification ([Wickham et al., 2022](#)).

All the publications reviewed so far formulated the HSCN optimization problem as a MILP, assuming linear relations of hydrogen production and utility consumption. Moreover, technologies integrating carbon capture were often modeled as single entities, ignoring important design and operating constraints. A more accurate representation of hydrogen production technologies is necessary not only to improve the quality of the optimal solutions, but to also provide a better understanding of the effects of these operating variables on the hydrogen supply chain. It is, therefore, the main objective of this work to present a framework to incorporate the rigorous modeling of hydrogen production technologies as part of a hydrogen strategic investment planning and supply chain optimization model. The problem is formulated as a multi-period MINLP and it is used to select the best technologies to produce, store, and transport hydrogen across large geographic envelopes and over extensive timeframes. In the temporal dimension, the model aims to produce an optimal long-term investment strategy to deploy and expand hydrogen production, storage, and primary distribution capacity. Purity requirements are very important and are addressed by considering two hydrogen purities: low (95 %) used by the industrial sector, and high (99.9 %+) used by the transportation sector. The contributions of this work are summarized below:

- 1 We use nonlinear reduced order models (ROMs) to better represent the hydrogen production and carbon capture processes for the optimal design of a Hydrogen Supply Chain Network (HSCN).
- 2 We model the process of carbon capture as a separate modeling construct using ROMs that account for individual mass balances for flue gas components and solvents.
- 3 We consider the production and distribution of hydrogen with different purity specifications for use in both transportation and/or industrial processes. The purification cost is a function of the product distribution, and it is obtained for each type of technology and the final use of hydrogen.
- 4 We finally present a detailed case study for the entire state of California that accounts for all the state's decarbonization mandates and optimally incorporates the time-varying emissions targets over a ten-year horizon ([CARB, 2018](#)).

The rest of the work is organized as follows. First, we present an overall description of the network topology as a superstructure model representation used to design the HSCN. Then, we describe the methodology used to build ROMs and to formulate of the overall model. [Section 4](#) highlights the modeling and optimization aspects of this work using the California hydrogen economy plan as a case study. [Section 5](#) discusses some aspects of the ROMs formulated and the results obtained from the case study. Finally, we summarize our conclusions in [Section 6](#).

2. Superstructure description

The proposed problem involves selecting the best long-term configuration of a hydrogen supply chain that maximizes profitability while meeting projected hydrogen demand and complying with carbon emission regulatory constraints. The model must select the best location and technologies to produce, store, and transport hydrogen with different modes and purities. [Fig. 1a](#) provides a high-level description of the five key modeling envelopes that constitute the entire hydrogen supply chain network. [Fig. 1b](#) provides a more detailed picture of the superstructure modeling approach for the hydrogen supply chain, including the specific process unit operations and the specific network interconnection among the six modeling blocks: primary hydrogen production facilities, low and high purification facilities, phase conditioning (i.e., compression and liquefaction), storage, transmission, and delivery. Here, we considered coal gasification (with and without carbon capture), steam methane reforming (with and without carbon capture), biomass gasification (with and without carbon capture), and

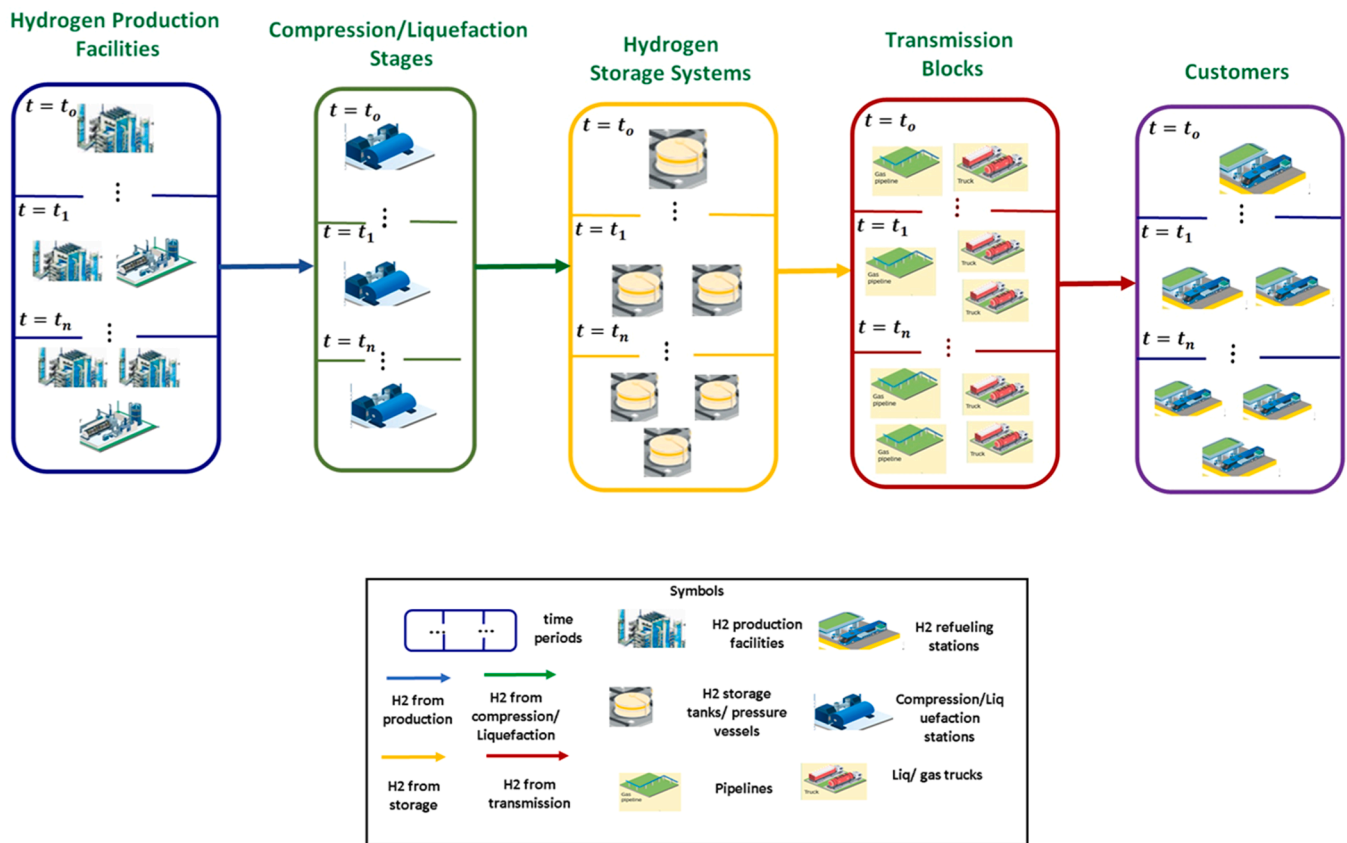


Fig. 1a. The conceptual layout of the hydrogen supply chain network.

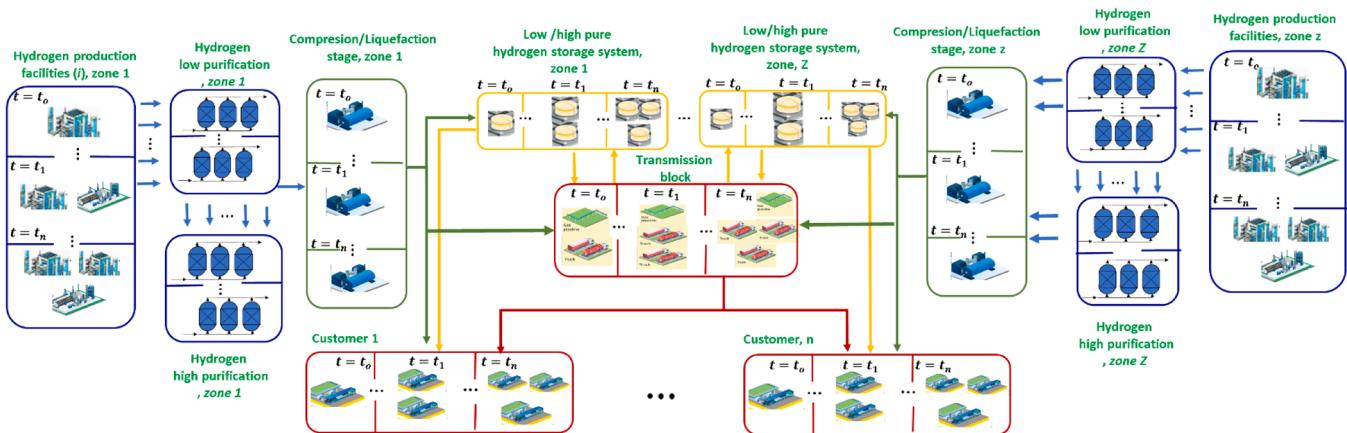


Fig. 1b. The detailed topology of the superstructure representation for the hydrogen supply chain network.

electrolysis as options to produce hydrogen. Long-term capacity expansion for each plant, ranging from small to centralized sizes, was also included as part of the formulation.

We assumed that the area of study can be divided into smaller zones based on the customer locations. The customers in each zone are clustered into two categories based on the hydrogen final use: automotive or industrial. For that purpose, we assumed two qualities of hydrogen to be supplied, i.e., low (used by the industrial sector) and high purity (consumed by the automotive sector). The former is obtained from the low purification block with 95 % as purity, while the latter requires an extra purification step to meet the specifications of fuel cell electric vehicles, i.e., 99.9 %.

Low- and high-purity hydrogen is sent to the conditioning section, where it is compressed or liquefied depending on its final disposition. For instance, hydrogen is compressed to 350 bars when stored as compressed gas, to 180 bars when transferred using gas trucks, and 100 bars when transferred using pipelines or liquefied when stored and transported as a liquid. After being conditioned, hydrogen is transferred (either imported or exported) to other zones by using gas and liquid trucks or pipelines. It is worth mentioning that the conditioning, storage, and transportation sections are modeled separately for both types of hydrogen, avoiding blending between the two product qualities. The transferred hydrogen can be stored, sent directly to the customer (in case of low purity hydrogen), or, in case of high purity hydrogen, sent to the

fueling stations located in the zone. In this regard, the model also considers the costs of installing new fueling stations used to supply the projected demand for high-purity hydrogen.

3. Model description

In this section, we present the methodology followed to construct reduced order models. Then, in Section 3.2, the main modeling equations and assumptions are described.

3.1. Methodology to develop a reduced order model

Reduced order models (ROMs) were constructed aimed at capturing the nonlinear behavior of the SMR and carbon capture technologies respectively. The proposed approach is presented in Fig. 2 and is described as follows. First, the main units of the process are modeled in Aspen HYSYS® (Aspen Technology, 2023a) based on rigorous first principles equations. Here, the overall performance of the process is studied through sensitivity analyses of the key operating variables. Then, multiple scenarios for those operating variables are created in Aspen multi-case® (Aspen Technology, 2023b) with the aim to solve multiple simulations in parallel and generate enough data to construct the ROM. This framework also requires the selection of independent and dependent variables. In this work, the independent variables were related with the main input streams of the process (i.e., Feedstock flowrate) and some operating variables (i.e., temperature and pressure). On the other hand, the dependent variables include the output streams of the process (i.e., products flowrate) and utilities consumption. A detailed description is shown in the next section. Once generated, the data is exported as a .json file to Aspen AI model builder® (AIMB). The

latter utilizes artificial intelligence principles to correlate the data gotten from the simulations and construct non-linear relationships (also called ROMs) between independent and dependent variables (Aspen Technology, 2023c). In this work, we used the linear regression method that incorporates polynomial terms. Once created, the ROM is deployed as an.XLP file containing the matrix structure and non-linear coefficient information that can be consumed by the Aspen technology proprietary MINLP solver called Aspen XSLP® (Varvarezos, 2008; Khor and Varvarezos, 2017). Finally, the optimization problem is solved in Aspen PIMS-AO®, as well as Aspen Unified PIMS®, the planning optimization and modeling frameworks most widely used in the petroleum and petrochemical industry to solve large MINLP problems including multi-period and multi-plant models (Aspen Technology, 2023d, 2023e).

3.1.1. Steam methane reforming

The modeling of this section was developed as a reduced-order model derived from rigorous simulation data using Aspen HYSYS (Aspen Technology, 2023a). The process consists of three sections: reformer reactor, high-and low-temperature water gas shift (WGS) reactors, and tail gas recycling and is described as follows. Natural gas and steam are sent to the reformer reactor and converted into syngas (a mixture of CO, CO₂, and H₂). Then, the stream is sent to the water gas shift reactors where the carbon monoxide and the remaining methane are converted into hydrogen plus CO₂. After that, the mixture is sent to a separation unit from which two streams are obtained: hydrogen and the remaining tail gasses. The tail gasses are sent to a combustion chamber to produce the steam required by the reformer reactor.

The amount of steam generated in the process highly depends on the tail gas stream composition. In this regard, two scenarios are possible:

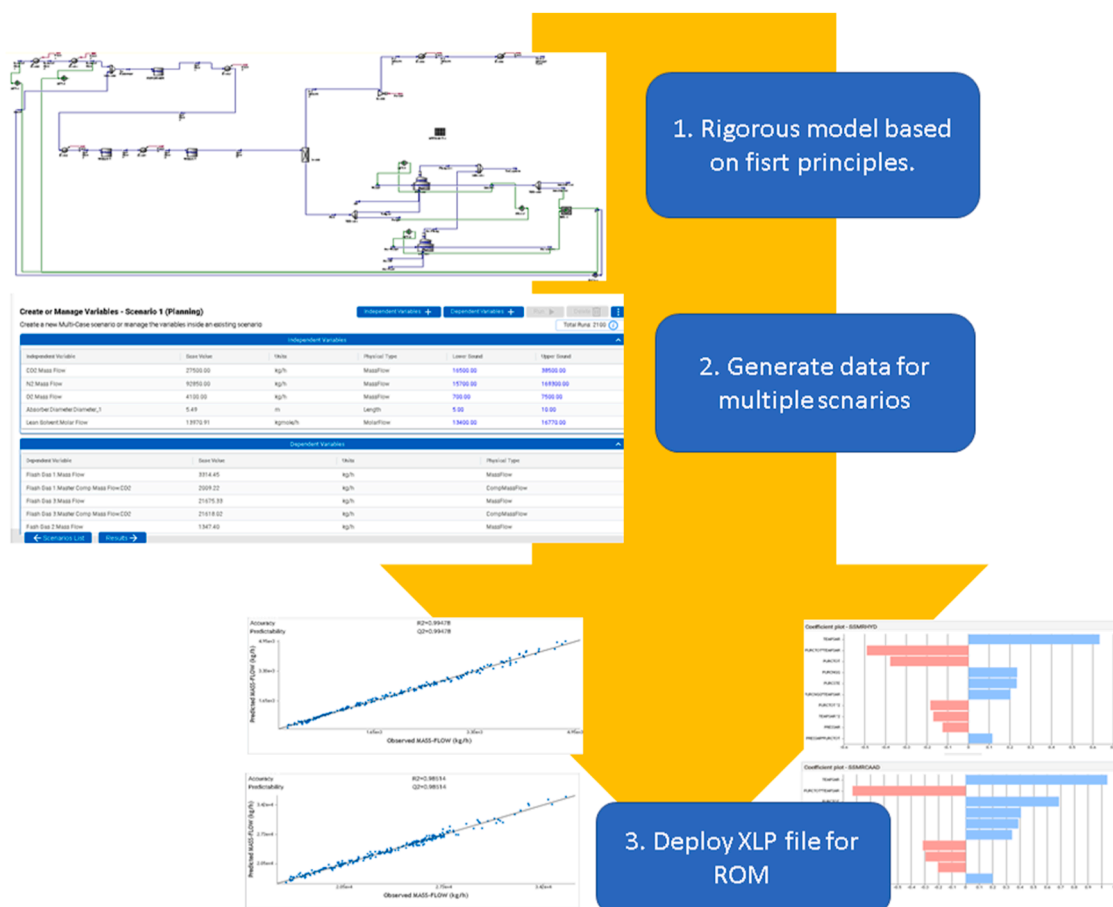


Fig. 2. Steps followed to construct reduced order models.

(a) the tail gasses stream can produce more steam than required, and (b) tail gasses are not enough to meet the steam requirements. For the first scenario, we considered the excess can be exported to other units; for the latter, we included a second combustion chamber that burns auxiliary natural gas to meet the steam needs.

3.1.2. Carbon capture process

In this process, a saturated flue gas stream, coming from the SMR process, is fed into an absorption column that uses DEPG as a solvent to remove the CO₂. Two output streams are obtained: sweet gas and rich-solvent streams. The first one is vented to the atmosphere, while the latter is sent to a recovery system consisting of flash drums. The solubility of CO₂ in DEPG solvent is highly dependent on the total pressure of the system. Therefore, the solvent recovery requires a set of flash drums that gradually reduce the pressure of the system to desorb the CO₂ (Ashkanani et al., 2020). The recovered solvent is recirculated to the absorption column, while the desorbed CO₂ is stored and sequestered.

The modeling of this section was formulated as a reduced order model. Here, the values generated by the ROM representing SMR (i.e., CO₂, N₂, and O₂ flowrates) are used as the inputs of the ROM for the CO₂ capture units. In this respect, the SMR process accounts for two sources of CO₂ emissions. The first one concerns the emissions coming from the reactor heating, while the second includes the CO₂ produced in the reactors and combustion chambers (see Fig. 3). We assumed that both streams can be combined and used as the input of the ROM for the CO₂ capture units. Hence, the model must determine the optimal fraction of the CO₂ to capture. The input stream is limited by constraints (1)–(15).

$$F^{in,M} = F^{h,s1} + F^{pi,s2} \quad (1)$$

$$F^{in,M} = \sum_{j \in J} F_j^{in,M} \quad (2)$$

$$F_j^{in,M} = F_j^{h,s1} + F_j^{p,s2} \quad \forall j \in J, \forall t \in T \quad (3)$$

$$F^{e,s1} = F^{h,s1} + F^{he,s1} \quad (4)$$

$$F^{e,s1} = \sum_{j \in J} F_j^{e,s1} \quad (5)$$

$$F_j^{e,s1} = F^{e,s1} x_j^{e,s1} \quad \forall j \in J, \forall t \in T \quad (6)$$

$$F_j^{h,s1} = F^{h,s1} x_j^{h,s1} \quad \forall j \in J, \forall t \in T \quad (7)$$

$$F_j^{he,s1} = F^{he,s1} x_j^{he,s1} \quad \forall j \in J, \forall t \in T \quad (8)$$

$$\sum_j x_j^{e,s1} \leq 1 \quad (9)$$

$$F^{p,s2} = F^{pi,s2} + F^{pe,s2} \quad (10)$$

$$F^{p,s2} = \sum_{j \in J} F_j^{p,s2} \quad (11)$$

$$F_j^{p,s2} = F^{p,s2} x_j^{p,s2} \quad \forall j \in J, \forall t \in T \quad (12)$$

$$F_j^{pi,s2} = F^{pi,s2} x_j^{pi,s2} \quad \forall j \in J, \forall t \in T \quad (13)$$

$$F_j^{pe,s2} = F^{pe,s2} x_j^{pe,s2} \quad \forall j \in J, \forall t \in T \quad (14)$$

$$\sum_j x_j^{p,s2} \leq 1 \quad (15)$$

3.2. Modeling of other production technologies

Electrolysis, coal, and biomass gasification were considered as alternative routes to produce hydrogen. To this end, the hydrogen production, CO₂ emissions, feedstocks, and utility consumption are calculated using yield-based models. The following equations were included.

$$H_{i,z,t}^{GEN} = F_{k,z,i,t} R_{i,k}^{HYD} \quad \forall i \in I, \forall z \in Z, \forall k \in K, \forall t \in T \quad (16)$$

$$KWH_{z,i,t} = F_{k,z,i,t} R_{i,k}^{KWH} \quad \forall i \in I, \forall z \in Z, \forall k \in K, \forall t \in T \quad (17)$$

$$CRD_{z,i,t} = F_{k,z,i,t} R_{i,k}^{CRD} \quad \forall i \in I, \forall z \in Z, \forall k \in K, \forall t \in T \quad (18)$$

where $R_{i,k}^{HYD}$, $R_{i,k}^{CRD}$, $R_{i,k}^{KWH}$ are parameters defined as the tons of H₂, CO₂ produced in unit i per tons of raw material k and the MWh of electricity required by unit i per tons of raw material k , respectively. $H_{i,z,t}^{GEN}$, $CRD_{z,i,t}$ and $KWH_{z,i,t}$ are the hydrogen produced, CO₂ emitted, and the electricity consumed in zone z by technology i at period t , respectively. $F_{k,z,i,t}$ denotes the raw material k fed to unit i at zone z .

3.3. Purification systems

Hydrogen purification was modeled as a separated section and consists of two groups of pressure swing adsorption (PSA) units. The first group, called low purification hydrogen, takes the hydrogen produced in the production block and removes some of the off-gasses (i.e., CH₄, CO, CO₂,) reaching a purity of 95 % (low-purity hydrogen). The second group is used to increase the purity of hydrogen from 95 % to 99 %, when required. The amount of off-gas to be removed in the low

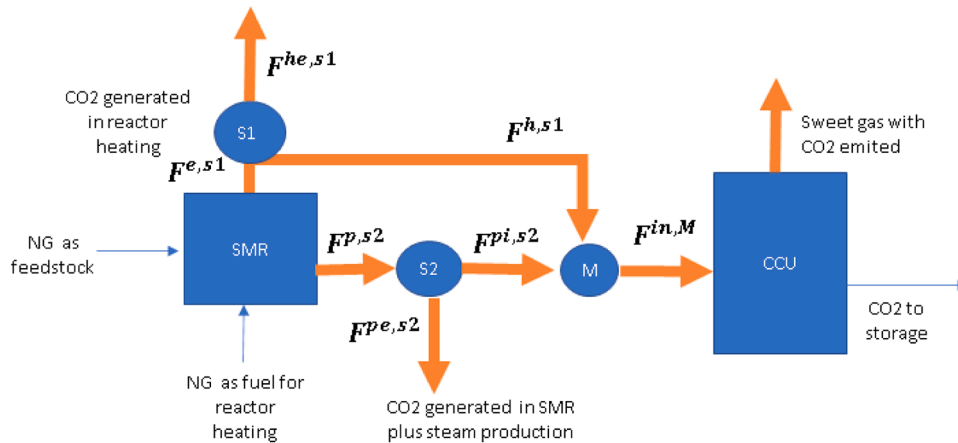


Fig. 3. Block diagram used to represent SMR emissions and carbon capture units.

Table 1

Assumed product distribution for each hydrogen production technology (Du et al., 2021).

	Coal Gasification	Biomass Gasification	Electrolysis
H ₂	30 %	30 %	99 %
CO ₂	40 %	35 %	–
CO	19.25 %	12 %	–
CH ₄	2 %	15 %	–
N ₂	0.75 %	1 %	–
H ₂ O	8 %	7 %	–

purification step is estimated based on the typical product distributions reported in literature for each type of technology, see Table 1. It is worth mentioning that the product distribution coming from SMR is not assumed but calculated as part of the ROM.

3.4. Supply chain equations

3.4.1. Mass balance constraints

The overall mass balance of any type of hydrogen f in each zone z and period t is stated as the summation of the hydrogen coming from the purification facilities and generated by technology i ($H_{i,f,z,t}^{PU}$) plus the amount of hydrogen imported from neighboring zones z' ($H_{f,z \rightarrow z',t}^{TRA}$), and the amount of hydrogen discharged by storage s ($H_{s,f,z,t}^{DIS}$) equals to the hydrogen demand ($D_{f,z,t}$), plus the amount of hydrogen exported to neighboring zones z' ($H_{f,z \rightarrow z',t}^{TRA}$), and the amount of hydrogen charged to storage s ($H_{s,f,z,t}^{DIS}$).

$$\sum_{i \in I} H_{i,f,z,t}^{PU} + \sum_{\forall z' \in Z, z \in Z} \sum_{l \in L} H_{f,z \rightarrow z',t}^{TRA} + \sum_{s \in S} H_{s,f,z,t}^{DIS} = D_{f,z,t} + \sum_{z \rightarrow z' \in ZZ} \sum_{l \in L} H_{f,z \rightarrow z',t}^{TRA} + \sum_{s \in S} H_{s,f,z,t}^{CHA} \quad \forall f \in F, \forall z \in Z, \forall t \in T \quad (19)$$

Eq. (20) states that the hydrogen generated in the production facilities is sent to the purification section, which produces low-and high-purity hydrogen.

$$H_{i,z,t}^{GEN} = \sum_{f \in F} H_{i,f,z,t}^{PU} \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (20)$$

3.4.2. Production capacity constraints

The capacity of each technology at period t ($Cap_{i,z,t}$) is calculated as the sum of the existing capacity at the previous period, plus the expanded capacity at period t ($Cexp_{i,z,t}^r$), minus the capacity retired at the same period ($Cre_{i,z,t}^r$).

$$Cap_{i,z,t} = Cap_{i,z,t-1} + \sum_r Cexp_{i,z,t}^r - \sum_{re} Cre_{i,z,t}^r \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (21)$$

The capacity expansion of each technology is conducted by installing new plants of different sizes. We considered plant sizes ranging from distributed to centralized. In this way, the capacity expansion is constrained as follows.

$$N_{i,z,t}^r Cmin_i^r \leq Cexp_{i,z,t}^r \leq Cmax_i^r N_{i,z,t}^r \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (22)$$

The number of plants installed at each period of time ($N_{i,z,t}^r$) is restricted by the following equation,

$$N_{i,z,t}^r \leq Nlmax * y_{i,z,t}^r \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (23)$$

where $y_{i,z,t}^r$ is a binary variable used to denote the installation of technology i at period t , and $Nlmax$ is the maximum number plants that can

be installed in one zone.

In a similar way, the retirement of plant is conducted follows,

$$N_{i,z,t}^r Cmin_i^r \leq Cre_{i,z,t}^r \leq Cmax_i^r N_{i,z,t}^r \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (24)$$

The number of plants retired at each period of time ($N_{i,z,t}^r$) is restricted by the following equation,

$$N_{i,z,t}^r \leq Nremax * y_{i,z,t}^r \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (25)$$

where $y_{i,z,t}^r$ is a binary variable denoting the retirement of technology i at period t , and $Nremax$ is the maximum number plants that can be retired at the same period. Additionally, Eq. (26) states that the installation and retirement of plants cannot be conducted at the same period of time,

$$y_{i,z,t}^r + y_{i,z,t}^r \leq 1 \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (26)$$

3.4.3. Storage systems constraints

Low- and high purity hydrogen can be stored as a compressed gas or as a liquid. The storage systems are modeled separately avoiding blending between both products. In this way, the amount of hydrogen stored in storage s at each period ($H_{s,f,z,t}^{STO}$) is calculated as the summation of the hydrogen stored at the previous period, plus the hydrogen charged coming from the purification system ($H_{s,f,z,t}^{CHA}$), minus the hydrogen discharged ($H_{s,f,z,t}^{DIS}$).

$$H_{s,f,z,t}^{STO} = H_{s,f,z,t-1}^{STO} + (H_{s,f,z,t}^{CHA} - H_{s,f,z,t}^{DIS}) \Delta t \quad \forall z \in Z, \forall s \in S, \forall f \in F, \forall t \in T \quad (27)$$

The capacity of each type of storage is bounded within its upper and lower limits as indicated by the following equation,

$$y_{s,f,z,t}^S Cmin_s^S \leq H_{s,f,z,t}^{STO} \leq Cmax_s^S y_{s,f,z,t}^S \quad \forall z \in Z, \forall s \in S, \forall f \in F, \forall t \in T \quad (28)$$

where $y_{s,f,z,t}^S$ indicates the installation of storage type s in zone z for product f . Note that capacity expansion of storage system was not considered in this model.

3.4.4. Transfer constraints

Hydrogen can be transferred from zone z to zone z' using pipelines, liquid, and gas trucks. The amount of hydrogen transferred using transport l ($H_{f,z \rightarrow z',t}^{TRA}$) is limited using Eq. (29).

$$y_{f,z \rightarrow z',t}^T H_{min}^{TRA} \leq H_{f,z \rightarrow z',t}^{TRA} \leq H_{max}^{TRA} y_{f,z \rightarrow z',t}^T \quad \forall f \in F, \forall z \rightarrow z' \in ZZ, \forall l \in L, \forall t \in T \quad (29)$$

where $y_{f,z \rightarrow z',t}^T$ denotes the existence of hydrogen transfer from zone z to zone z' .

Hydrogen crossed importation and exportation between two zones are not allowed to occur at the same time period. This item is indicated by,

$$y_{f,z \rightarrow z',t}^T + \sum_{l \in L} \sum_{f \in F} y_{f,z' \rightarrow z,t}^T \leq 1 \quad \forall f \in F, \forall z \rightarrow z' \in ZZ, \forall l \in L, \forall t \in T \quad (30)$$

The existence of any hydrogen transfer associated with a storage system (i.e., hydrogen transferred from storage and other zones) must be subjected to its installation. For that purpose, we define the following

constraint,

$$y_{f,z \rightarrow z',t}^T - y_{s,f,z,t}^S \leq 0 \quad \forall f \in F, \forall z \rightarrow z' (z = s) \in ZZ', \forall l \in L, \forall t \in T \quad (31)$$

Note that Eq. (32) applies only for variables $y_{f,z \rightarrow z',t}^T$ concerning storage transfers.

3.4.5. Refueling stations constraints

Refueling stations were considered as the final user of the high-purity hydrogen. Their total capacity at each period t ($Cap_{z,t}^{rs}$) is evaluated as the summation of the existing capacity at the previous period, plus the expanded capacity at period t ($Cexp_{z,t}^{rs}$).

$$Cap_{z,t}^{rs} = Cap_{z,t-1}^{rs} + Cexp_{z,t}^{rs} \quad \forall z \in Z, \forall t \in T \quad (32)$$

The capacity expansion of refueling stations is conducted by installing new stations. To this end, the following equation is included,

$$N_{z,t}^{rs} Cmin_z^r \leq Cexp_{z,t}^r \leq Cmax_z^r N_{z,t}^{rs} \quad \forall z \in Z, \forall t \in T \quad (33)$$

where $N_{z,t}^{rs}$ is an integer variable denoting the number of new refueling stations installed in zone z at period t .

3.5. Investment and fixed capital costs

3.5.1. Production facilities

The total investment cost of each type of production technology is calculated based on the installation and capacity expansions made at each period of time. The following type of equation was used,

$$CI_{i,z,t} = \sum_{r \in R} KA_{i,t}^r Cexp_{i,z,t}^r + \sum_{r \in R} KB_{i,t}^r N_{i,z,t}^r \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (34)$$

where $A_{i,t}^r$ and $B_{i,t}^r$ are constant whose values depend on the type and size of technology to be expanded, K denotes the depreciation factor. Then, the fixed capital cost ($PC_{i,z,t}$) is calculated as the depreciated value of the capital investment. Note, that once a unit installed, its depreciation cost must be added to the annual costs the rest of the project life or when its lifespan ends. Here, we assumed that the lifespan of production units is larger than the project life. Therefore, the fixed capital cost at each period is calculated as the accumulated sum of the depreciated values of previous and new expansions.

$$PC_{i,z,t} = \sum_t CI_{i,z,t} \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (35)$$

3.5.2. Purification facilities

The capital investment cost of purification is function of the amount of off-gas produced in the hydrogen production block and is calculated as follows (Demirhan et al., 2019),

$$CIPur_{i,z,t} = 7.17 (S_{off-gas,i,z,t})^{0.65} \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (36)$$

Off-gas production is constrained by the following Eqn.

$$S_{off-gas,i,z,t} \geq S_{off-gas,i,z,t-1} \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (37)$$

Finally, the capital cost for the high purification section was assumed as 35.96 \$/kWh_{H2} (Wickham et al., 2022). In this way, the total fixed capital cost of the purification system is given by Eq. (38).

$$PurC_{i,f,z,t} = K (CIPur_{i,z,t} + GH_{i,f,z,t}^{PU}) \quad \forall i \in I, \forall f \in F, \forall z \in Z, \forall t \in T \quad (38)$$

where K is the depreciation factor, and G is a conversion factor used to convert the unit investment cost of the high purification section.

3.5.3. Transportation cost

The fixed capital cost of transportation is estimated based on the number of pipelines, gas, and liquid trucks required to transfer hydrogen from one zone to another as stated by Eq. (39).

$$TC_{f,l,t} = KC_{l,t}^{TRA} (N_{f,l,t}^{TRA}) \quad \forall f \in F, \forall l \in L, \forall t \in T \quad (39)$$

where $C_{l,t}^{TRA}$ is the unit investment cost required for each type of transportation mean, $N_{f,l,t}^{TRA}$ is an integer variable denoting the total number of trucks/pipelines installed or purchased at each period of time. It is calculated as the summation of the average number of trucks transferring hydrogen between two zones ($N_{f,l,z \rightarrow z',t}$),

$$N_{f,l,t}^{TRA} = \sum_{z \rightarrow z' \in ZZ} N_{f,l,z \rightarrow z',t} \quad \forall f \in F, \forall l \in L, \forall t \in T \quad (40)$$

The number of trucks available at each period, is restricted by the following equation,

$$N_{f,l,t}^{TRA} \geq N_{f,l,t-1}^{TRA} \quad \forall f \in F, \forall l \in L, \forall t \in T \quad (41)$$

Note that low-and high-purity hydrogen are transported using separate means. Then, the variable $N_{f,l,z \rightarrow z',t}$ is estimated using the correlation proposed by Guillén-Gosálbez et al. (2010).

$$N_{f,l,z \rightarrow z',t} = \frac{2(ZZ') (H_{f,z \rightarrow z',t}^{TRA})}{SP_l * Cap_l} \quad \forall f \in F, \forall l \in L, \forall z \rightarrow z' \in ZZ', \forall t \in T \quad (42)$$

where ZZ' is the average distance between two zones in mi, SP_l is the average speed of trucks, measured in mi/day, Cap_l is the capacity of each truck l .

3.5.4. Refueling stations

The capital cost of the refueling stations is estimated based on the new stations installed at each time-period.

$$RSC_{z,t} = KC_{z,t}^{RS} (N_{z,t}^{rs}) \quad \forall z \in Z, \forall t \in T \quad (43)$$

Where $C_{z,t}^{RS}$ is the unit investment cost of a refueling station, $N_{z,t}^{rs}$ is the number of stations installed in zone z , which are constrained by the following equation:

$$N_{z,t}^{rs} = \sum_t N_{z,t}^{rs} \quad \forall z \in Z, \forall t \in T \quad (44)$$

3.6. Operating cost

3.6.1. Production and purification sections

Operating cost of production and purification sections considers the costs of, raw materials, electricity, carbon taxes, and auxiliary fuel when required (SMR). Those are expressed as follows,

$$OpC_{i,z,t} = C_{i,k}^{RW} F_{k,z,i,t} + C_i^{kwh} KWH_{z,i,t} + C_i^{CO2} CRD_{z,i,t} + C_i^{AF} AF_{z,i=SMR,t} \quad \forall i \in I, \forall z \in Z, \forall t \in T \quad (45)$$

$$OpCost_{i,f,t}^{pur} = C_i^{kwh} R_f^{KWH} H_{i,f,z,t}^{PU} \quad \forall i \in I, \forall f \in F, \forall z \in Z, \forall t \in T \quad (46)$$

where $C_{i,k}^{RW}$, C_i^{kwh} , C_i^{CO2} , C_i^{AF} are the unit costs of raw materials, electricity, carbon taxes, and auxiliary fuel, respectively, which are time dependent.

3.6.2. Storage systems

The total operating cost of each storage systems is expressed as the summation of the electricity required to compress or liquified hydrogen, plus the holding costs. Here, we assumed 11 MWh/tonH₂, and 1.95 MW/tonH₂ as the energy required to compressed (350 bar) and liquified hydrogen, respectively (He et al., 2021).

$$OpCost_{f,s,z,t}^{STO} = C_i^{kwh} R_s^{KWH} H_{s,f,z,t}^{CHA} + Hcost_{i,s,f,z,t}^{STO} \quad \forall f \in F, \forall s \in S, \forall z \in Z, \forall t \in T \quad (47)$$

3.6.3. Transportation means

Operating costs of liquid and gas trucks are divided into two items. Namely, the cost of the electricity required to compress (at 180 bar) and liquified gas, and the fuel cost required to transport hydrogen between zones.

$$OpCost_{f,l,t}^{TR} = C_t^{kwh} R_l^{KWH} H_{f,z \rightarrow z',t}^{TRA} + FC_{f,l,z \rightarrow z',t}$$

$$\forall f \in F, \forall z \rightarrow z' \in ZZ', \forall l(L=LT, GT) \in L, \forall t \in T \quad (48)$$

where $FC_{f,l,z \rightarrow z',t}$ is the fuel cost, which is calculated as follows (Guillén-Gosálbez et al., 2010),

$$FC_{f,l,z \rightarrow z',t} = \frac{2(FP)(ZZ') \left(H_{f,z \rightarrow z',t}^{TRA} \right)}{FE * Cap_l} \quad \forall f \in F, \forall z \rightarrow z' \in ZZ', \forall l(L=LT, GT) \in L, \forall t \in T \quad (49)$$

where ZZ' is the average distance between two zones in mi, FP , is the fuel price (\$/gal), and FE is the fuel economy (mi/gal*truck). On the other hand, operating cost of pipelines are estimated using the correlations proposed by He et al. (2021),

$$OpCost_{f,l,t}^{TR} = C_t^{kwh} \left(H_{f,z \rightarrow z',t}^{TRA} (ZZ') + 1 \right) \quad \forall f \in F, \forall z \rightarrow z' \in ZZ', \forall l(L=PIP) \in L, \forall t \in T \quad (50)$$

Note that, the above equation estimates the compression cost when hydrogen is transported by a pipeline based on the distance ZZ' between two zones.

3.7. Objective function

We aim at maximizing the net present value (NPV) of the hydrogen network, which is evaluated as the summation of the discounted cash flows associated with hydrogen revenues ($HSELL_{f,z,t}$), capital and operating costs related with production ($PC_{i,z,t}$ and $OpC_{i,z,t}$), purification costs ($PurC_{i,f,t}$ and $OpCost_{i,f,t}^{pur}$), storage costs ($SC_{f,s,z,t}$ and $OpCost_{f,s,z,t}^{STO}$), transportation costs ($TC_{f,l,t}$ and $OpCost_{f,l,t}^{TR}$), and delivery costs ($RSC_{z,t}$).

$$NPV = \sum_{z,f,t} \frac{HSELL_{f,z,t}}{(1+r)^t} - \sum_{z,i,t} \frac{PC_{i,z,t}}{(1+r)^t} - \sum_{z,i,t} \frac{OpC_{i,z,t}}{(1+r)^t} - \sum_{i,f,z,t} \frac{PurC_{i,f,z,t}}{(1+r)^t} - \sum_{i,f,z,t} \frac{OpCost_{i,f,z,t}^{pur}}{(1+r)^t} - \sum_{f,l,t} \frac{TC_{f,l,t}}{(1+r)^t} - \sum_{f,l,t} \frac{OpCost_{f,l,t}^{TR}}{(1+r)^t} - \sum_{f,s,z,t} \frac{SC_{f,s,z,t}}{(1+r)^t} - \sum_{f,s,z,t} \frac{OpCost_{f,s,z,t}^{STO}}{(1+r)^t} - \sum_{z,t} \frac{RSC_{z,t}}{(1+r)^t} \quad (51)$$

where r is the interest rate and the term $HSELL_{f,z,t}$ considers the revenues of low and high-purity hydrogen selling.

4. Case study

The model presented here was used to optimize the strategic planning investment for developing the hydrogen supply chain in the state of California. One hydrogen consumer in California is the transportation sector. According to the California Air Resources Board (CARB, 2021), the state of California currently has 67 fueling stations installed, with an average capacity of 250 kg/day, that supply the hydrogen required by over 10,000 fuel-cell electric vehicles (FCEV). As part of the same efforts to decarbonize the sector, the current legislation (Assembly Bill 32, 2006), mandates the decrease of carbon emissions to the 1990's level and increase the use of renewables from 50 % to 100 % to produce energy by the end of this decade (CARB, 2018). The other large hydrogen consumer in the state, is the refinery sector with a total demand of 2100 Mt/day, which is supplied on-site or by gas companies through SMR process. In this work, we use the proposed approach to supply the demand of both customers assuming they require different qualities of hydrogen, i.e., high-purity hydrogen for fueling stations and low-purity for refineries.

For modeling purposes, we have lumped the hydrogen demand in four zones that roughly correspond to the cities having the largest capacity of fueling stations, namely, San Francisco, Sacramento, San Diego, and Los Angeles (see Fig. 4a and b, AFDC, 2022). Installation of storage was not included because of the coarse time discretization used. We also considered to supply the hydrogen required by the refineries located at those cities based on information taken from Gilani and Sanchez (2020). The estimated hydrogen demand for each zone is shown in Table S10. Here, we considered a time horizon of ten years from 2023 to 2032. Carbon emissions for SMR, coal, and biomass gasification were estimated as the CO₂ that is directly emitted by the process technologies. As for electrolysis, we assumed the emission factors of electricity calculated by EPA (2022). Those emissions factors were then adjusted to

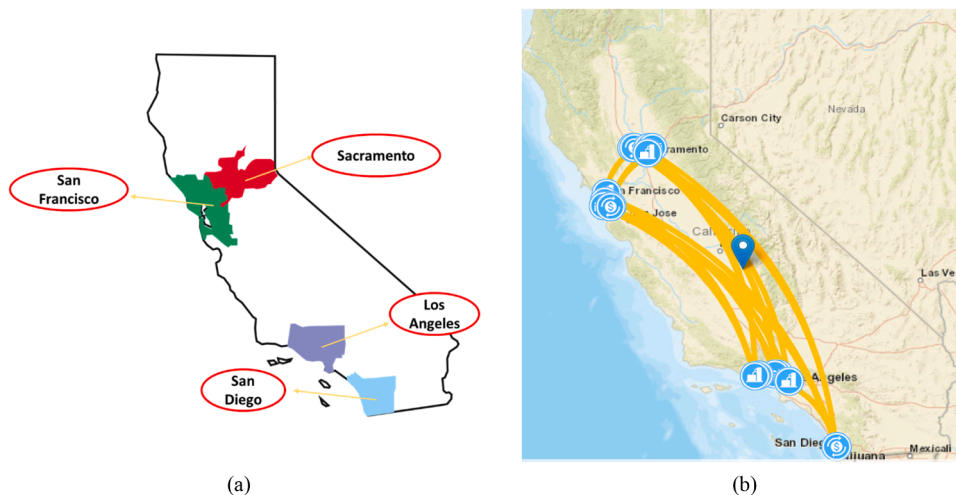


Fig. 4. a. Cities of the state of California accounting for the largest hydrogen fueling capacity b. Map visualization of possible transfers between cities of the state of California in Aspen Unified PIMS® (Aspen Technology, 2023e).

account for the expected increase of renewables power production share from 50 % to 100 % by 2032. For that purpose, we considered 10 years of time horizon accounting for a gradual increase on the renewables share, i.e., from the current 50 % to 70 % by 2027, and 100 % by 2032. Similarly, the model includes CO₂ emissions constraints that gradually constrain the total amount of CO₂ generated by the related processes. The model was developed in Aspen PIMS-AO (Varvarezos et al., 2001; Varvarezos 2008). It consists of 22,760 continuous variables, 27,796 equations, and 3600 integer variables, and it was solved in 480 s as CPU time using the solver Aspen XSLP (Varvarezos et al., 2001; Varvarezos 2008; Khor and Varvarezos 2017) to solve the MINLP problem.

5. Results

In this section, a brief discussion on the ROMs obtained for the SMR and CCU processes is presented. Then, we discuss the hydrogen supply chain design and capacity deployment for the case study from two perspectives. First, we solve the problem with no restrictions on the carbon emission. Later, a second solution is presented including the carbon emission constraints inscribed in the legislation of the state of California.

5.1. Discussion on the reduced order models

As discussed, ROMs for steam methane reforming and carbon capture were developed through AI model builder. They consist of polynomial functions taking care of relevant variables that might affect the performance of the process. In this section, we discussed the non-linear behavior of the process which is revealed in the coefficients and parity plots.

5.1.1. Steam methane reformer process

Fig. 5 shows the coefficient plots for the dependent variables studied in the SMR process. Note that the y-axis indicates the polynomial terms included in the ROM, while the x-axis shows the coefficient value of each term. Here, one can observe that the linear terms are predominant in the ROMs and influenced by the total feed variable (SSMRINS). The latter captures all the input streams of the process. Namely, natural gas as feedstock, and the air and water used to produce steam. Then, we added natural gas and steam inputs as two separate variables aiming at

studying their individual contribution. Overall, the natural gas feed (SSMRNGG) accounts for the most predominant effect over the products flowrates. On the other hand, the non-linearities of the process are mainly triggered by the reformer temperature (SSMRTEM) appearing in the quadratic and bilinear terms. The coefficient plots show that a temperature increase favors the hydrogen production, but it also leads to higher CO₂ emissions. The last statement makes sense since most of the SMR and WGS reactions are endothermic. Therefore, a major decision of the model is to find the trade-off between these two conflicting variables. Unlike reformer temperature, reformer pressure does not show a significant effect on the dependent variables studied.

5.1.2. Carbon capture process

The dependent variables studied in the carbon capture process correspond with the CO₂ content in the output streams. Namely, sweet-gas stream and the streams coming from the flash drums. In this case, the following remarks can be extracted from the coefficient plots shown in Fig. 6. First, an increase in N₂ and O₂ content in the flue gas input leads to an increase of CO₂ in the sweet gas stream. A higher N₂ and O₂ content reduces the partial pressure of CO₂ in the flue gas, and, consequently, its solubility in DEPG. Second, the non-linearities of the process are driven by three variables, i.e., the flowrates of solvent, CO₂ and N₂ fed into the absorption column and affect the overall performance of the recovery system. Coefficient plots show that an increase in the flowrate of solvent and nitrogen leads to a decrease in the total amount of CO₂ obtained in the first and the second flash drums. This item is explained by the high solubility of CO₂ at the operating pressures of both flash drums (172 kPa and 101.3 kPa). In fact, it was observed that over 60 % of the total CO₂ content in DEPG is desorbed in the third flash drum operating at 27.38 kPa.

5.2. Hydrogen supply chain long-term design

5.2.1. Scenario A: maximum NPV with no carbon emissions constraints

The first case corresponds to the maximization of the net present value with no constraints on carbon emissions. In this solution, the NPV yielded a value of \$3744.2 MM for the planning horizon, while the total carbon emissions at the last period accounted for 4327 Mt/day. The optimal configuration of the hydrogen supply chain is shown in Fig. 7, and it includes steam methane reforming and electrolysis to produce

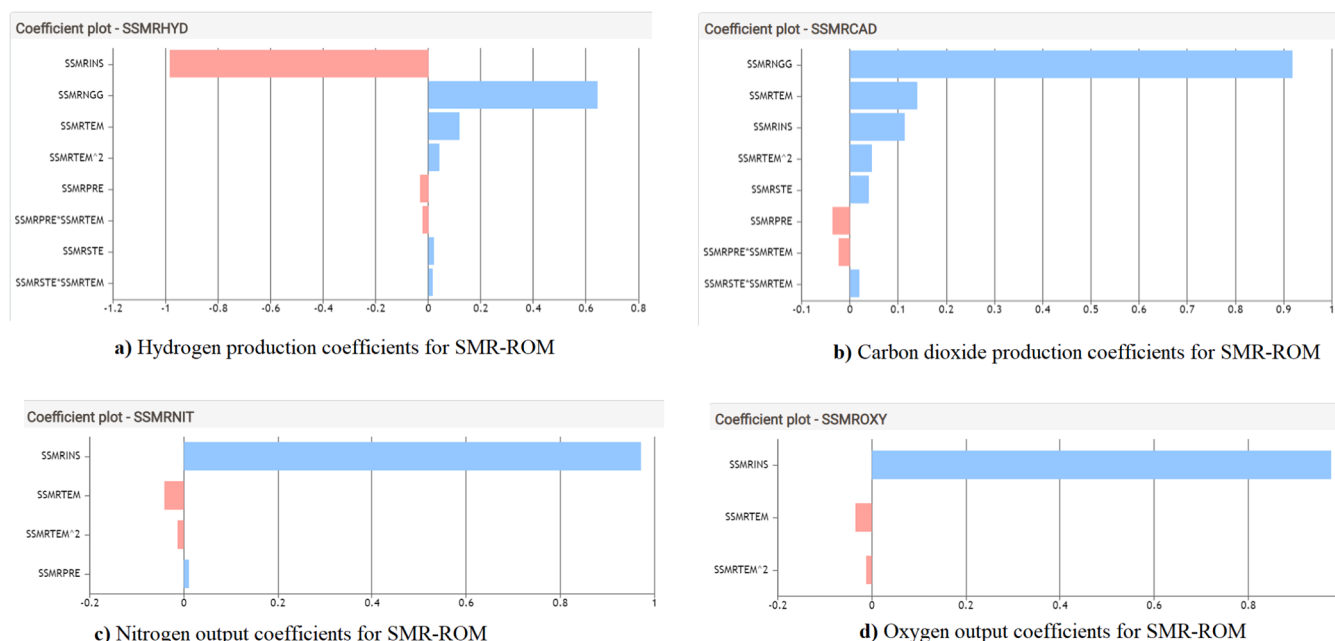


Fig. 5. Coefficient plots for dependent variables for ROM representing SMR.

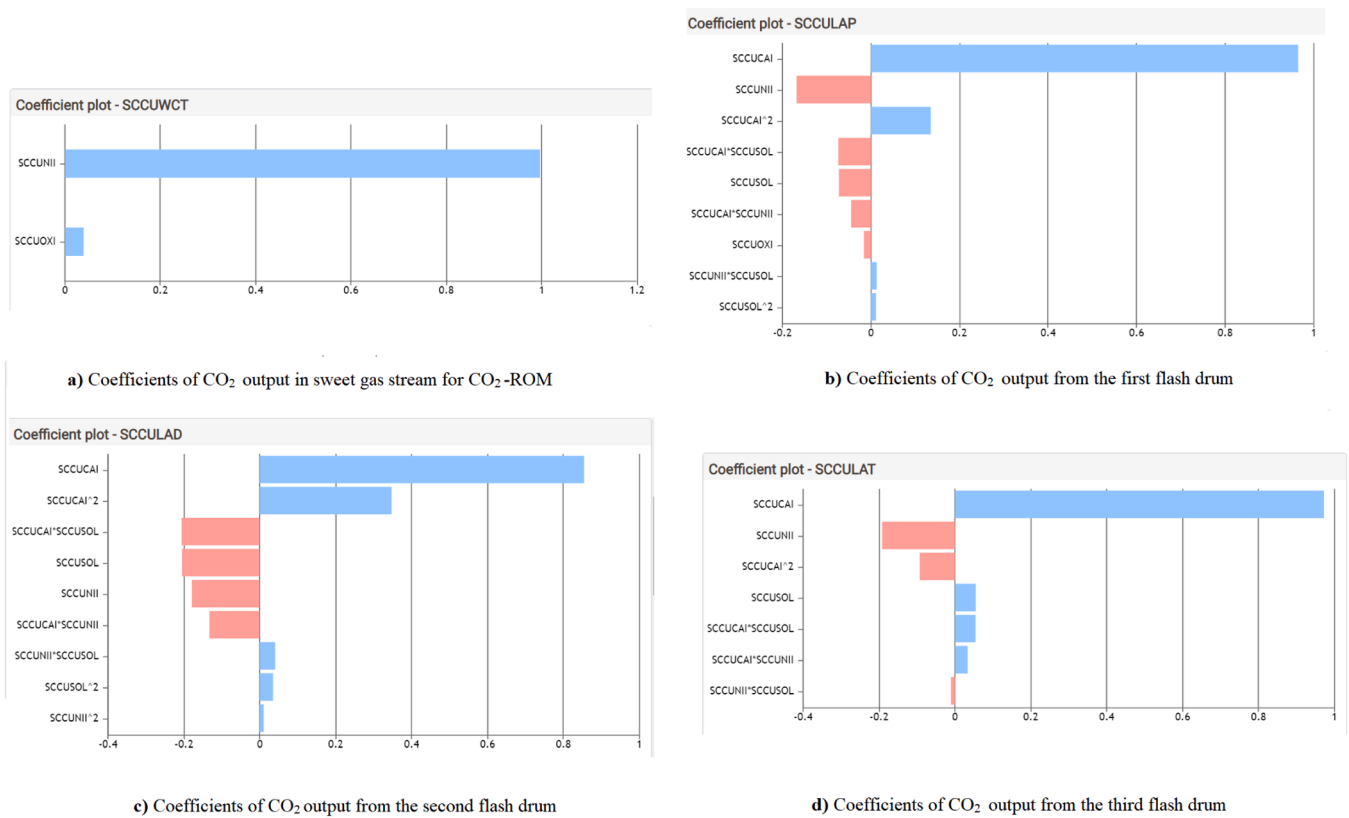


Fig. 6. Coefficient plots of dependent variables for ROM for the CO₂ capture units.

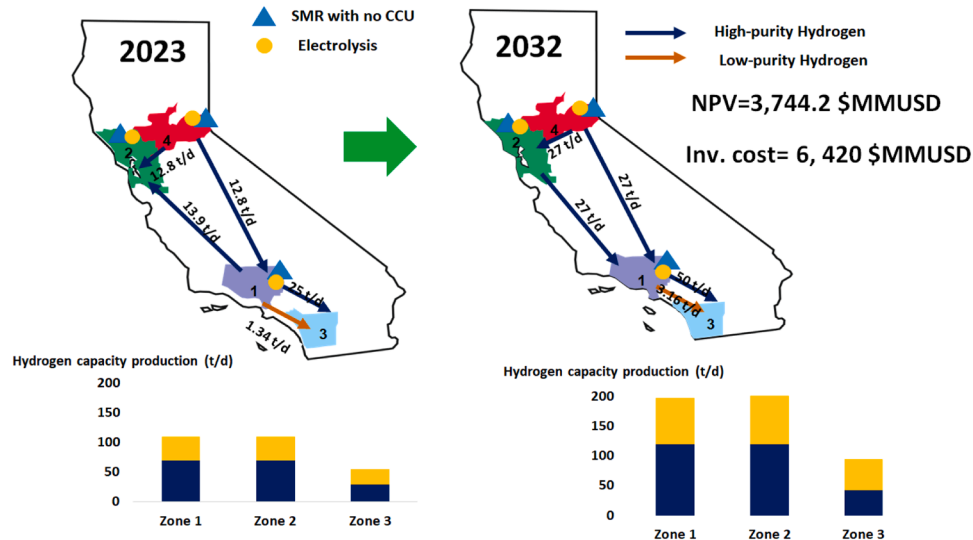


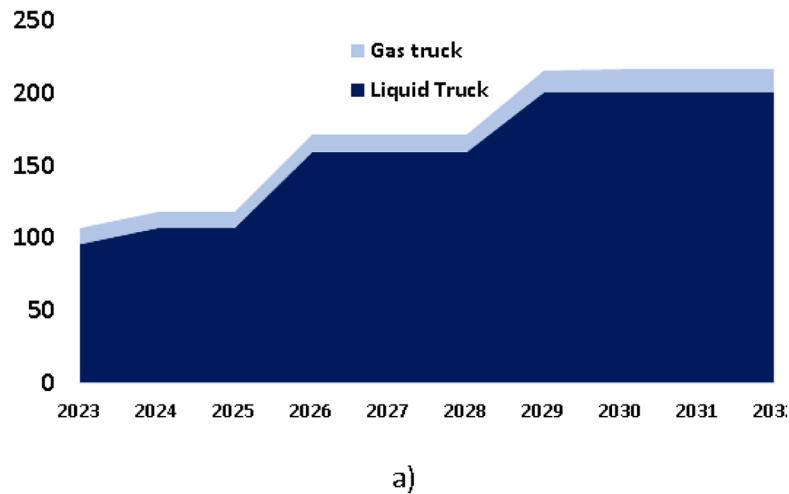
Fig. 7. Optimal HSCN design for Scenario A: maximization of NPV with no carbon constraints.

hydrogen. At the beginning of the project, the SMR technology accounts for 170 t/d as total capacity, which is expanded to 182 t/d in 2026, and to 283 t/d in 2032. Results showed that the total demand for low-purity hydrogen is supplied by the SMR facilities, while high-purity hydrogen is produced by electrolysis. To this end, the model selects an initial capacity of 102.7 t/d, which is gradually expanded through periods 2026 and 2032 to 150 and 207 t/d, respectively. It can also be seen that the high-purity hydrogen is the main product to be transferred between the zones. In this respect, zone 4 becomes the major supplier of zones 1 and 2, which are the largest hydrogen consumers. The optimal Scenario Also determines zone 1 as the major supplier of zone 3. Hydrogen is

transferred using gas and liquid trucks (see Fig. 8a). It was observed that the gas trucks are used when small amounts of hydrogen are required to be transferred in a short distance, i.e., low-purity hydrogen transferred between zones 1 and 3. The increasing demands for high-purity hydrogen also had an impact of the fueling stations installed in each zone whose capacity is expanded by 50 % at the last period (see Fig. 8b).

An additional analysis revealed that at the break-even point (i.e., NPV=0), the CO₂ emissions over the whole-time horizon are 90 % lower than the ones calculated in the base case. Furthermore, when 1 % of loss in profitability is considered in a Pareto optimality sense, the total carbon emissions can be reduced by 10.8 %. In this respect, two strategies

Hydrogen capacity Transportation (t/d)



Fueling stations capacity (t/d)

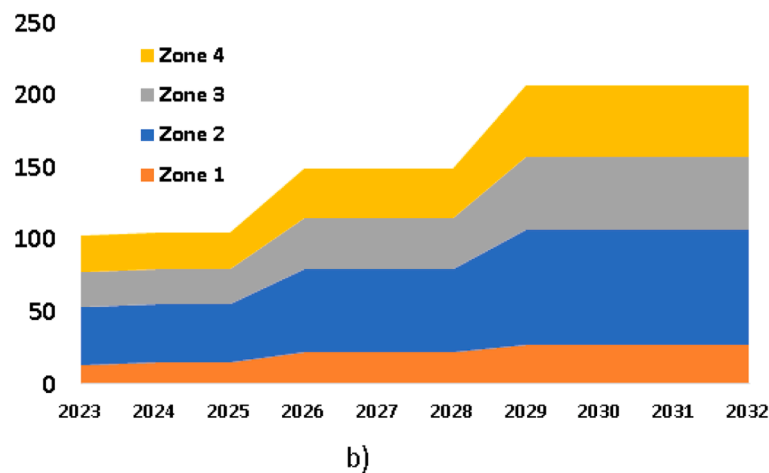


Fig. 8. Capacity expansion for Scenario A: (a) Hydrogen transport, and (b) fueling stations.

Table 2

Optimization results for different carbon tax rates and regulatory restrictions.

Case	Cumulative Emissions (10 yrs) (kTon)	Carbon emission reduction (%)	Net Present value (\$MM)	Investment Cost (\$MM)
No CO ₂ restrictions				
Carbon Tax 0 \$/t (Scenario A)	11,881	0 %	3744	6420
Carbon Tax 50 \$/t	10,580	11 %	3333	5611
Carbon Tax 100 \$/t	10,229	14 %	2962	5423
Carbon Tax 150 \$/t	9620	19 %	2592	5131
CO ₂ restrictions				
Carbon Tax 0 \$/t (Scenario B)	3223	73 %	1363	6908

could be followed to favor the reduction of CO₂ emissions. The first one considers a carbon tax aiming at penalizing the production of CO₂. In the second option, specific goals (based on policies) for carbon reduction can be included as part of the model formulation. The latter case is discussed in Section 5.2.2. As for the former, we conducted three additional case studies including 50, 100, and 150 \$/t as a carbon tax. Results revealed that the total carbon emissions over the whole-time horizon were reduced by 11 % when 50 \$/t was considered. The highest reduction was reached when 150 \$/t was used, yielding 9.6 Mt, 19 % lower than the base case. Nevertheless, the profitability of the system decreased by 30 %. The reader is referred to Table 2 for further detail.

5.2.2. Scenario B: maximum NPV under carbon emissions policy. gradual CO₂ emissions reduction

The second solution corresponds to an infrastructure that maximizes the NPV of the network, while meeting the CO₂ emissions reduction policy goals set by the state of California. For that purpose, we set the goal to reduce the emissions calculated in Scenario A at the last period by assuming a progressive reduction of 80 %, 85 %, and 90 % for the years 2026, 2029, and 2032, respectively. The model provided a solution with a NPV of \$1363 MM and a total investment cost of 6,907 \$MMUSD (just under \$7 billion). This estimation is reasonable when compared the expectations of the California government that

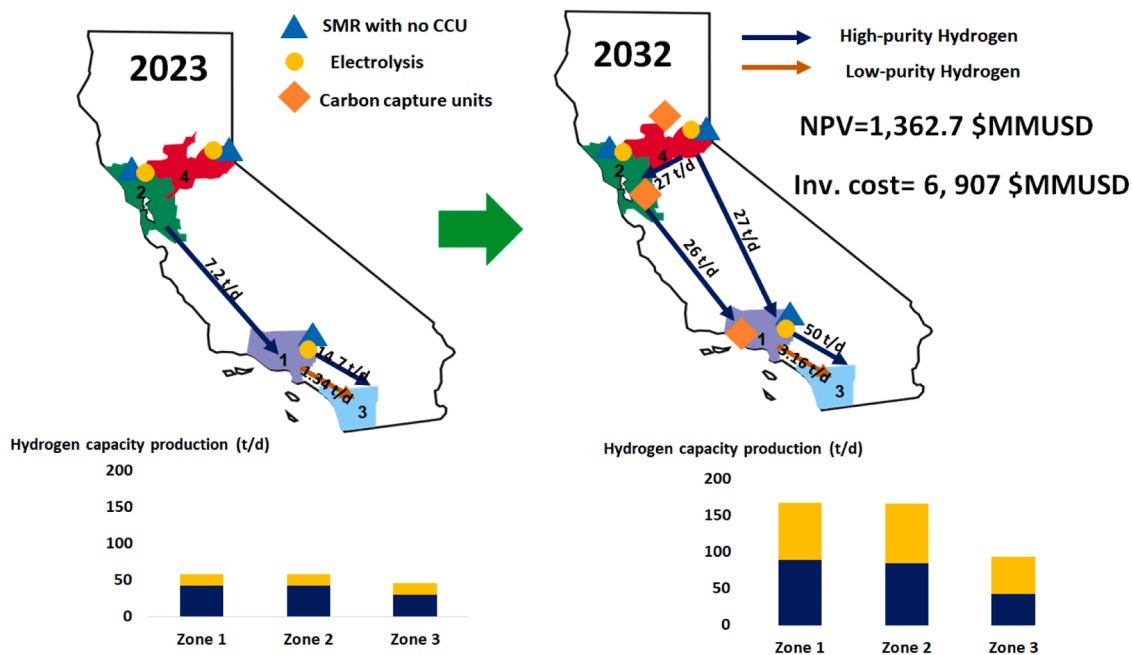
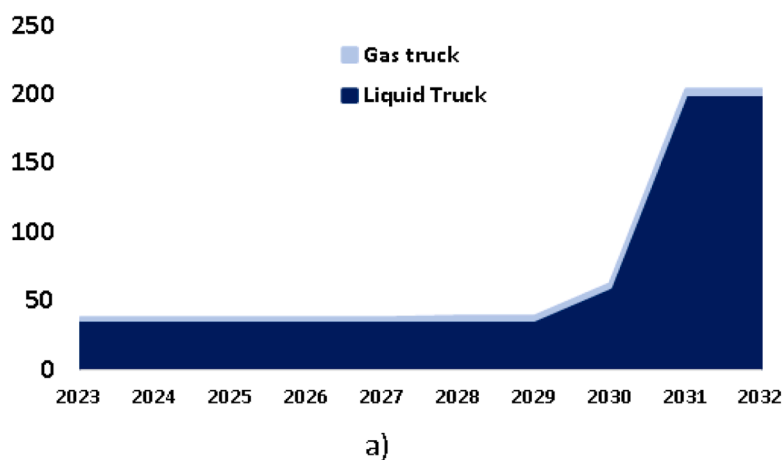


Fig. 9. Optimal HSCN design for Scenario B: maximization of NPV including carbon emissions constraints.

Hydrogen capacity Transportation (t/d)



Fueling stations capacity (t/d)

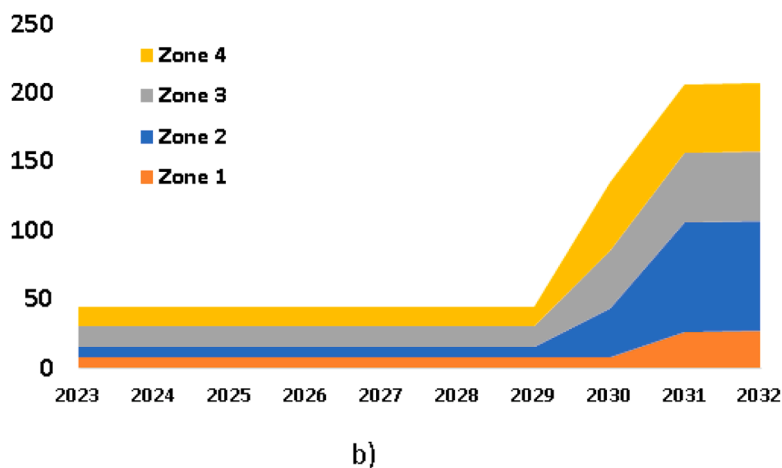


Fig. 10. Capacity expansion for Scenario B: (a) Hydrogen transport, and (b) fueling stations.

committed, in 2021, the allocation of \$8 billion dollars for the development of hydrogen infrastructure in at least four regions along the state (GO-Biz, 2022). Moreover, the total emissions at the last period were calculated as 400 MT/day, being 90 % lower than the ones calculated in Scenario A at year 2032. Here, the investment decisions were clearly driven by the CO₂ constraints at each period. Results showed that, at the beginning of the project, the model installed smaller capacities of SMR and electrolysis (116 and 46 MT/day, respectively), which are used to mainly supply the local demand of low- and high-purity hydrogen, reducing the hydrogen transferred between the zones. In 2026, when the CO₂ restrictions increase to 80 %, the capacity expansion of SMR was conducted with the installation of carbon capture units yielding 217 MT/day as total capacity (22 % lower than Scenario A). (Fig. 9)

The higher renewable penetration in power grid expected by 2032 (reflected as lowering emissions factors of electricity) and the increasing emissions constraints at the last three periods prompted the deployment of electrolysis to produce high-purity hydrogen, which is first expanded from to 135 MT/day, in 2030, and to 207 MT/day in 2032. This item was also reflected in the installed capacity of fueling stations. The model determined the installation of 64 new fueling stations at the beginning of the project yielding 44 MT/day as total capacity (see Fig. 10). According to the last projection of the California energy commission, the state could reach this capacity by 2026. However, it is also mentioned that this estimation might change based on the growth rate of fuel-cells electric vehicles penetration, among other factors that are usually uncertain (Baronas and Belinda, 2021). The total number of fueling stations installed at the end of the project was 712 yielding 207 MT/day as total capacity. The use of liquid trucks to transport high purity hydrogen was also preferred over gas trucks and pipelines.

6. Conclusions

A mixed-integer non-linear model was developed for the strategic investment planning related to the design and long-term capacity expansion of hydrogen supply chains. The best configuration of the HSCN is determined through a superstructure that maximizes the profitability of the network, while meeting the projected hydrogen demand. It includes six blocks: production, purification, conditioning, storage, transport, and delivery. We also presented a methodology to build reduced order models aimed at capturing the business-important non-linearities of the technologies involved in the production block. The methodology uses data extracted from rigorous simulations models to construct non-linear relationships between the dependent and independent variables. In this respect, results also showed that the necessity of including non-linearities for the SMR is triggered by the temperature of the reformer reactor. Coefficient plots revealed that a temperature increase favors the hydrogen production, but it also leads to higher CO₂ emissions. As for the carbon capture process, we observed that the solvent recovery step was mainly affected by the total amount of nitrogen content and the solvent flowrate required to absorb CO₂. An increase in the flowrate of solvent and nitrogen leads to a decrease in the total amount of CO₂ obtained in the flash drums used to recover the solvent. These models were included in the HSCN model used to solve a case study aimed at the development of hydrogen infrastructure for the state of California. The technoeconomic analysis showed that the use of SMR and electrolysis were the preferred technologies to produce hydrogen. In particular, SMR plants are used to satisfy the demand for low-purity hydrogen, while electrolysis plants supply the high-purity hydrogen for transportation. The optimal solution also calls for liquid trucks to transfer hydrogen between the zones and gas trucks for smaller quantities and short distances. In this respect, low-purity hydrogen was mainly transferred using compressed hydrogen gas trucks, while high-purity hydrogen transportation utilized primarily liquid hydrogen trucks. The increasing CO₂ emissions constraints strongly influenced the decisions on how and when the infrastructure was deployed. Results showed that the expansion of SMR was conducted with the installation

of carbon units. As the CO₂ constraint became more restrictive, the capacity expansion of electrolysis was favored over SMR, largely because of the expected increase of renewable share in power grid by the end of the decade. The total number of fueling stations capacity in periods 2023–2029 was similar to the one projected by the California energy commission. Moreover, it was also shown that the expansion of electrolysis prompted the installation of new fueling stations by 2032. Under the latter scheme, the optimizer was able to reduce the carbon emissions at the last period by 90 % with respect to the base case. Results also showed that when 1 % of loss in profitability is allowed, the total carbon emissions can be reduced by 10.8 %. As for the carbon tax, optimization results revealed that even with a modest penalty, carbon emissions can be reduced by 11 % and up to 20 % compared to a 73 % carbon reduction resulting from the regulatory framework in place and without any explicit carbon tax (as can be seen in Table 2).

Finally, it is worth noting, that the optimal total investment cost results are consistent with the stated estimates from the state of California. In fact, the optimal policy approach, developed here, yields a lower total investment (\$6.9B versus \$8B allocation, (GO-Biz, 2022)), while satisfying all the projected hydrogen demands and the emission constraints from the underlying regulatory framework. Furthermore, this investment not only reduces emissions (by 90 % in 2032), but it is also cash-flow positive with a discounted cash flow of \$1.4B. We expect that such modeling and optimization frameworks will greatly assist the accelerated and economical achievement of worldwide emission reduction targets set by both corporate and government entities.

CRedit authorship contribution statement

Salvador I. Pérez-Uresti: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Gustavo Gallardo:** Methodology, Software, Validation, Formal analysis, Investigation. **Dimitrios K. Varvarezos:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Salvador Israel Perez Uresti reports a relationship with Aspen Technology Inc that includes: employment.

Data availability

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.compchemeng.2023.108412.

References

- Agnolucci, P., Akgul, O., McDowall, W., Papageorgiou, L.G., 2013. The importance of economies of scale, transport costs and demand patterns in optimising hydrogen fuelling infrastructure: an exploration with SHIPMod (Spatial hydrogen infrastructure planning model). *Int. J. Hydrog. Energy* 38 (26), 11189–11201. <https://doi.org/10.1016/j.ijhydene.2013.06.071>.
- Almansoori, A., Shah, N., 2006. Design and operation of a future hydrogen supply chain: snapshot model. *Chem. Eng. Res. Des.* 84 (6), 423–438. <https://doi.org/10.1205/cherd.05193>.
- Almansoori, A., Shah, N., 2009. Design and operation of a future hydrogen supply chain: multi-period model. *Int. J. Hydrog. Energy* 34 (19), 7883–7897. <https://doi.org/10.1016/j.ijhydene.2009.07.109>.

- Almansoori, A., Shah, N., 2012. Design and operation of a stochastic hydrogen supply chain network under demand uncertainty. *Int. J. Hydrog. Energy* 37 (5), 3965–3977. <https://doi.org/10.1016/j.ijhydene.2011.11.091>.
- Ashkanani, H.E., Wang, R., Shi, W., Siefert, N.S., Thompson, R.L., Smith, K., ..., Morsi, B. I., 2020. Levelized cost of CO₂ captured using five physical solvents in pre-combustion applications. *Int. J. Greenh. Gas Control* 101, 103135. <https://doi.org/10.1016/j.jggc.2020.103135>.
- Aspen Technology. 2023c. Aspen Hybrid models. Combine the capabilities of first principles models, domain expertise and AI to expand the boundaries of what is possible for the process industry. Available at: <https://www.aspentech.com/en/solutions/aspen-hybrid-models>. (Accessed on January 2023).
- Aspen Technology. 2023a. Aspen HYSYS. Maximize safety, sustainability and profits by optimizing the entire site in one environment using industry-validated simulation and time-saving workflows. Available at: <https://www.aspentech.com/en/products/engineering/aspen-hysys>. (Accessed on January 2023).
- Aspen Technology. 2023b. Aspen multi-cases. Achieve a superior design by rapidly analyzing many process simulation cases in parallel – leveraging highperformance computing, machine learning, data analysis and visualization tools. Available at: <https://www.aspentech.com/en/products/engineering/aspen-multi-case>. (Accessed on January 2023).
- Aspen Technology. 2023d. Aspen PIMS-AO (Process industry modeling system—Advanced optimization). Optimize feedstocks, product slates, capital expansions and more with Aspen PIMS, used at more than 400 refineries and olefins plants worldwide. Available at: <https://www.aspentech.com/en/products/msc/aspen-pims>. (Accessed on January 2023).
- Aspen Technology. 2023d. Aspen Unified. Available at: <https://www.aspentech.com/en/resources/brochure/aspen-unified>. (Accessed on January 2023).
- Baronas, J., Belinda C. 2021. Joint agency staff report on assembly bill 8: 2021 annual assessment of time and cost needed to attain 100 hydrogen refueling stations in California. California energy commission and California air resources board. Publication Number: CEC-600-2021-040. Available at: <https://www.energy.ca.gov/sites/default/files/2021-12/CEC-600-2021-040.pdf>. (Accessed on September 2022).
- Barthélémy, H., Weber, M., Barbier, F., 2017. Hydrogen storage: recent improvements and industrial perspectives. *Int. J. Hydrog. Energy* 42 (11), 7254–7262. <https://doi.org/10.1016/j.ijhydene.2016.03.178>.
- Bique, A.O., Maia, L.K., Grossmann, I.E., Zondervan, E., 2021. Design of hydrogen supply chains under demand uncertainty—a case study of passenger transport in Germany. *Phys. Sci. Rev.* <https://doi.org/10.1515/psr-2020-0052>.
- California air resources board (CARB). 2018. AB 32 global warming solutions act 2006. Available at: <https://ww2.arb.ca.gov/es/node/2235>. (Accessed on September 2022).
- California air resources board (CARB). 2021. 2021 annual evaluation of fuel cell electric vehicle deployment and hydrogen fuel station network development. Available at: https://ww2.arb.ca.gov/sites/default/files/2021-09/2021_AB-8_FINAL.pdf. (Accessed on September 2022).
- Chiyoda corporation. 2020. Chiyoda's Hydrogen Supply Chain Buissiness. The world's first global hydrogen supply chain demonstration project. Available at: <https://www.chiyodacorp.com/en/service/spera-hydrogen/>. (Accessed November 2022).
- Davis, W., Martin, M., 2014. Optimal year-round operation for methane production from CO₂ and water using wind energy. *Energy* 69, 497–505. <https://doi.org/10.1016/j.jclepro.2014.05.077>.
- Demirhan, C.D., Tso, W.W., Powell, J.B., Pistikopoulos, E.N., 2019. Sustainable ammonia production through process synthesis and global optimization. *AIChE J.* 65 (7), e16498. <https://doi.org/10.1002/aic.16498>.
- Du, Z., Liu, C., Zhai, J., Guo, X., Xiong, Y., Su, W., He, G., 2021. A review of hydrogen purification technologies for fuel cell vehicles. *Catalysts* 11 (3), 393. <https://doi.org/10.3390/catal11030393>.
- Gilani, H.R.; Sanchez, D.L. 2020. Introduction to the hydrogen market. Available at: https://bof.fire.ca.gov/media/10190/introduction-to-the-hydrogen-market-in-california-draft-for-comment_ada.pdf. (Accessed on January 2023).
- Guerra, O.J., Eichman, J., Kurtz, J., Hodge, B.M., 2019. Cost competitiveness of electrolytic hydrogen. *Joule* 3 (10), 2425–2443. <https://doi.org/10.1016/j.joule.2019.07.006>.
- Guillén-Gosálbez, G., Mele, F.D., Grossmann, I.E., 2010. A bi-criterion optimization approach for the design and planning of hydrogen supply chains for vehicle use. *AIChE J.* 56 (3), 650–667. <https://doi.org/10.1002/aic.12024>.
- He, G., Mallapragada, D.S., Bose, A., Heuberger, C.F., Genç, E., 2021. Hydrogen supply chain planning with flexible transmission and storage scheduling. *IEEE Trans. Sustain. Energy* 12 (3), 1730–1740. <https://doi.org/10.1109/TSTE.2021.3064015>.
- Hydrogen council. 2021. Hydrogen insights. A perspective on hydrogen investment, market development, and cost competitiveness. Available at: <https://hydrogencouncil.com/wp-content/uploads/2021/02/Hydrogen-Insights-2021-Report.pdf>.
- Iberdrola. 2022. Iberdrola commissions the largest green hydrogen plant for industrial use in Europe. Available at: <https://www.iberdrola.com/about-us/what-we-do/green-hydrogen/puertollano-green-hydrogen-plant>. (Accessed on January 2023).
- International energy agency (IEA). 2019. The future of hydrogen, Sizing today's opportunities. Available at: <https://www.iea.org/reports/the-future-of-hydrogen>.
- International energy agency (IEA). 2021. Global hydrogen review 2021. Available at: <https://iea.blob.core.windows.net/assets/5bd46d7b-906a-4429-abda-e9c507a62341/GlobalHydrogenReview2021.pdf>. (Accessed on January 2023).
- International Renewable Energy Agency (IRENA), 2019. Hydrogen: a Renewable Energy Perspective. Available at: <https://www.irena.org/publications/2019/Sep/Hydrogen-A-renewable-energy-perspective>. (Accessed on January 2023).
- Khor, C.S., Varvarezos, D., 2017. Petroleum refinery optimization. *Optim. Eng.* 18 (4), 943–989. <https://doi.org/10.1007/s11081-016-9338-x>.
- Linde. 2022. Linde to increase green hydrogen production in the United States. Available at: <https://www.linde.com/news-media/press-releases/2022/linde-to-increase-green-hydrogen-production-in-the-united-states>. (Accessed on January 2023).
- Melaina, M.W., Antonia, O., & Penev, M. 2013. Blending hydrogen into natural gas pipeline networks: a review of key issues. Available at: <https://www.h2knowledgecentre.com/content/researchpaper1511>. (Accessed on December 2022).
- Mosca, L., Jimenez, J.A.M., Wassie, S.A., Gallucci, F., Palo, E., Colozzi, M., Galdieri, G., 2020. Process design for green hydrogen production. *Int. J. Hydrog. Energy* 45 (12), 7266–7277. <https://doi.org/10.1016/j.ijhydene.2019.08.206>.
- National Grid gas system operator. 2021. Gas Future Operability planning (GFoP) report. Hydrogen blends in the NTS. A teorical exploration . Available at: <https://www.nationalgrid.com/gas-transmission/document/137506/download>. (Accessed on October 2022).
- Nunes, P., Oliveira, F., Hamacher, S., Almansoori, A., 2015. Design of a hydrogen supply chain with uncertainty. *Int. J. Hydrog. Energy* 40 (46), 16408–16418. <https://doi.org/10.1016/j.ijhydene.2015.10.015>.
- Penchev, M.; Lim, T., Todd, M., Lever, O., Lever, E., Mathaudhu, S., Martinez-Morales, A. & Raju, A.S.K. 2022. Hydrogen blending impacts study final report. Available at: <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M493/K760/493760600.PDF>. (Accessed on December 2022).
- REPHYNE. 2021. Shell starts up Europe's largest PEM green hydrogen electrolyser. Available at: <https://www.refhyne.eu/shell-starts-up-europes-largest-pem-green-hydrogen-electrolyser/#:~:text=The%20Rheinland%20electrolyser%20will%20use,10%20help%20decarbonise%20other%20industries>. (Accessed on November 2022).
- Reuß, M., Grube, T., Robinius, M., Stolten, D., 2019a. A hydrogen supply chain with spatial resolution: comparative analysis of infrastructure technologies in Germany. *Appl. Energy* 247, 438–453. <https://doi.org/10.1016/j.apenergy.2019.04.064>.
- Reuß, M., Welder, L., Thürauf, J., Linßen, J., Grube, T., Schewe, L., Robinius, M., 2019b. Modeling hydrogen networks for future energy systems: a comparison of linear and nonlinear approaches. *Int. J. Hydrog. Energy* 44 (60), 32136–32150. <https://doi.org/10.1016/j.ijhydene.2019.10.080>.
- Sabio, N., Gadalla, M., Guillén-Gosálbez, G., Jiménez, L., 2010. Strategic planning with risk control of hydrogen supply chains for vehicle use under uncertainty in operating costs: a case study of Spain. *Int. J. Hydrog. Energy* 35 (13), 6836–6852. <https://doi.org/10.1016/j.ijhydene.2010.04.010>.
- Sadler, D., Cargill, A., Crowther, M., Rennie, A., Watt, J., Burton, S., & Haines, M. 2016. H21 Leeds city gate. Leeds City Gate, Northern Gas Networks, Wales and West Utilities, 1325. Available at: <https://www.northerngasnetworks.co.uk/wp-content/uploads/2017/04/H21-Report-Interactive-PDF-July-2016.compressed.pdf>. (Accessed on December 2022).
- Samsatli, S., Samsatli, N.J., 2019. The role of renewable hydrogen and inter-seasonal storage in decarbonising heat—Comprehensive optimisation of future renewable energy value chains. *Appl. Energy* 233, 854–893. <https://doi.org/10.1016/j.apenergy.2018.09.159>.
- Samsatli, S., Staffell, I., Samsatli, N.J., 2016. Optimal design and operation of integrated wind-hydrogen-electricity networks for decarbonising the domestic transport sector in Great Britain. *Int. J. Hydrog. Energy* 41 (1), 447–475. <https://doi.org/10.1016/j.ijhydene.2015.10.032>.
- Shell. 2022. Shell to start building Europe's largest renewable hydrogen plant. Available at: <https://www.shell.com/media/news-and-media-releases/2022/shell-to-start-building-europes-largest-renewable-hydrogen-plant.html>. (Accessed on January 2023).
- Tarkowski, R., 2019. Underground hydrogen storage: characteristics and prospects. *Renew. Sust. Energy Rev.* 105, 86–94. <https://doi.org/10.1016/j.rser.2019.01.051>.
- The governor's office of business and economic development (GO-Biz), 2022. California's hydrogen goals. Available at: <https://business.ca.gov/industries/hydrogen/>. (Accessed on January 2023).
- U.S. Environmental Protection Agency (EPA), 2022. GHG emission factor hub. Available at: <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>. (Accessed on October 2022).
- US department of Engery. Alternative fuel data center (AFDC). 2022. Hydrogen fueling station locations. Available at: https://afdc.energy.gov/fuels/hydrogen_locations.html#/find/nearest?fuel=HY. (Accessed on September 2022).
- Varvarezos, D.K., Kunt, T., & Paules, G.E. 2001. Using other optimization technologies in PIMS. *PIMS-Users Conference Proceedings, October 2001*.
- Varvarezos, D.K., 2008. Optimal solution-range analysis in production planning: refinery feedstock selection. *Ind. Eng. Chem. Res.* 47 (21), 8282–8285. <https://doi.org/10.1021/ie800078e>.
- Wickham, D., Hawkes, A., Jalil-Vega, F., 2022. Hydrogen supply chain optimisation for the transport sector—Focus on hydrogen purity and purification requirements. *Appl. Energy* 305, 117740. <https://doi.org/10.1016/j.apenergy.2021.117740>.
- Yang, C., Ogden, J., 2007. Determining the lowest-cost hydrogen delivery mode. *Int. J. Hydrog. Energy* 32 (2), 268–286. <https://doi.org/10.1016/j.ijhydene.2006.05.009>.
- Yu, M., Wang, K., Vredenburg, H., 2021. Insights into low-carbon hydrogen production methods: green, blue and aqua hydrogen. *Int. J. Hydrog. Energy* 46 (41), 21261–21273. <https://doi.org/10.1016/j.ijhydene.2021.04.016>.